Nutrition and Health

Goal: Increase understanding on nutrition in the context of community health

Specific objectives:

- Describe nutrition and dietary requirements in our lives
- Describe the implications of malnutrition in health and development

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Nutrition - Nutrition is the intake of food, considered in relation to the body's dietary needs. Good nutrition is an adequate, well balanced diet combined with regular physical activity. It is the cornerstone of good health. Poor nutrition can lead to reduced immunity, increased susceptibility to disease, impaired physical and mental development, and reduced productivity. Eating well is important for everyone's health and wellbeing and is vital for preventing chronic disease in later life.

Community nutrition – is the nutrition in the context of community. In order for nutrition to have impact on human health and well-being, it must be applied to human populations through community and population based intervention and education programs.

Malnutrition - Malnutrition is the condition that develops when the body does not get the right amount of the vitamins, minerals, and other nutrients it needs to maintain healthy tissues and organ function. It is a cause and consequence of ill health across many groups; however, older people tend to be more vulnerable to this condition, which can severely impact on health outcomes and quality of life.

Basic nutrition - Basic nutrition is the food intake, dietary nutrients required during our phases of life

- *Food intake* – An average person takes at least 3 meals in a day (breakfast, lunch and supper). People tend to have supplementary feeds in between the main meals. Children are required to be fed every time according to their demand. Food intake is influenced by income, appetite, exposure to food and social status.
- *Dietary nutrients* – Our body requires essential dietary nutrients for its normal function. The nutrient requirements are: Carbohydrates, Proteins, Water, Vitamins, and Minerals. Essential nutrient are substance that must be obtained from the diet because the body cannot make it in sufficient quantity to meet its needs. In establishing dietary requirements the goal is to determine what amount is high enough, if ingested routinely, to prevent impairment of health even if intake becomes inadequate for a short period
- *Nutrition during phases of life* – This is the nutrition required during: Pregnancy and early childhood; Childhood and adolescent; School going age; Adults; and Older age.
    - *Pregnancy and early childhood* - Starting early with breastfeeding and providing good nutrition for children gives the best start to a healthy life. Good nutrition in pregnancy is essential for keeping a developing baby and mother healthy. Breastfeeding is the natural way to feed babies. Breastfeeding is extremely important for the health outcomes of infants and their mother. Not breastfeeding is associated with adverse health outcomes for both mothers and infants, ranging from infectious illness to chronic diseases later in life. Solids should be introduced at around six months of age when the infant is developmentally ready. For young children, a healthy diet is required for optimal growth and development. It can also help to improve concentration, fight illnesses, and prevent long term health complications.

- *Children and Adolescents* - Children and adolescents need sufficient foods to grow and develop normally. Growth should be checked regularly for young children. Physical activity is important for children and adolescents. Many infants suffer and die because of ignorance, meanness of parents and inability to understand the infant communication language

- *School aged children* - Childhood is an important period when children develop the knowledge, skills and behaviors for lifelong health and wellbeing. We know that education and health are inseparable and that a healthy child learns well.

- *Adults* - Healthy eating during adulthood is vital for effective functioning of body systems and immunity, maintaining a healthy weight, and preventing and managing chronic conditions such as diabetes and heart disease.

- *Older people* - Good nutrition is important for older people to help promote independence and to reduce illness and premature death. Maintaining a healthy lifestyle, including good nutrition, plays an important role in ensuring older people can live well at home for longer.

Components of community nutrition

Community nutrition entails the following components:

- *Surveillance of food chain* – This is assessing food for possible infestation with germs, toxic. It's also called food contamination monitoring
- *Nutritional epidemiology* - A medical research that studies the relationship between nutrition and health. It is the on effects of dietary intake and nutritional status on health of individual or community. Multiple nutrients work together in the body, with differing impacts on different systems and under differing environmental conditions.
- *Clinical nutrition and diabetes* - The number of patients with diabetes is growing rapidly. Involves assessing the effectiveness of *clinical nutrition* in diabetes care and outcomes
- *Nutrition education* **-** Is any combination of educational strategies, accompanied by environmental supports, designed to facilitate voluntary adoption of food choices and nutrition-related behaviors conducive to health and well-being. Nutrition education is delivered through multiple venues and involves activities at the individual, community, and policy levels
- *Food and nutrition policy* – It make healthy food more available; Disclose the calorie content of restaurant foods; Teach people how to make healthy eating easier. There are 3 existing nutrition policies and programs, like: Nutrition Facts labels on packaged foods; Nutrition standards for school lunches; Regulation of food additives.
- *Food industry and standard agency* - is a complex, global collective of diverse businesses that supply much of the food and food energy consumed by the world population e.g. UNILIVER. The Food Standards Agency is concerned with the whole food industry – from farming, food production and distribution, to retail and catering. It addresses food safety issues at every stage of the chain, providing information and guidance on best practice and legal requirements.
- *Mass catering* - Food and drink supplied by a caterer to large numbers of customers at the same time

Malnutrition

- Individuals are malnourished, or suffer from under nutrition if their diet does not provide them with adequate calories and protein for maintenance and growth, or they cannot fully utilize the food they eat due to illness.
- Growth failure malnutrition, as the name suggests, is the failure of an individual to grow as expected in stature or weight, according to his/her age and gender. Growth failure malnutrition can take three serious types:
    - Acute malnutrition or wasting arises out of sudden, drastic weight loss. It leads to three clinical malnutrition types:
        - *Marasmus*: this occurs when body fat and tissues degenerate at an alarming rate to compensate for the lack of nutrients. As a result, the body's internal processes begin to slow down alarmingly fast as does the activity of the immune system.
        - *Kwashiorkor*: this is characterized by bilateral pitting oedema (fluid retention) in the legs and feet. As a result, the under-nourished child may actually look plump.
        - *Marasmic-kwashiorkor*: This is characterized by both severe wasting and oedema.
    - Chronic malnutrition or stunting happens over a long period of time and has more long-lasting consequences. It begins before birth due to poor maternal health and leads to stunted growth in an otherwise normally proportioned child. Poor breast feeding, infections and lack of availability of proper nutrients are the main causes behind it. Stunting is dangerous because it becomes irreversible after an age. Therefore, it becomes important to nip it in the bud by providing proper medical treatment to pregnant women and young girls.
    - Micronutrient malnutrition implies a moderate to severe lack of Vitamins A, B, C and D, Calcium, Folate, Iodine, Iron, Zinc and Selenium. These vitamins and minerals are of utmost importance in various body processes and their deficiency can make an otherwise healthy person malnourished:
        - Iron deficiency causes anaemia, poor brain development and cardiac functioning.
        - Iodine deficiency leads to impaired thyroid functioning and mental retardation.
        - Vitamin D deficiency causes rickets and other bone development related disorders.
        - Selenium deficiency leads to poor cardiac function, weak immunity and osteoarthritis.
        - Vitamin A deficiency is a cause of poor vision, bone development and immunity.
        - Vitamin B12 deficiency leads to nerve degeneration and poor RBC formation
        - Folate or vitamin B9 deficiency causes slow growth and anemia
        - Zinc deficiency can cause poor immunity, sensory perception and anemia.

- Signs and symptoms of malnutrition- A symptom is something the patient feels and reports, while a sign is something other people, such as the doctor detect. For example, pain may be a symptom while a rash may be a sign. The general signs and symptoms include:
    - Loss of fat (adipose tissue)
    - Breathing difficulties, a higher risk of respiratory failure
    - Depression
    - Higher risk of complications after surgery and lower healing hence longer recovery
    - Higher risk of hypothermia - abnormally feeling cold and low body temperature
    - The total number of some types of white blood cells falls; consequently, the immune system is weakened, increasing the risk of infections.
    - Lower sex drive and problems with fertility
    - Reduced muscle and tissue mass
    - Tiredness, fatigue, apathy and irritability
    - In more severe cases:
        - Skin may become thin, dry, inelastic, pale, and cold
        - Eventually, as fat in the face is lost, the cheeks look hollow and the eyes sunken
        - Hair becomes dry and sparse, falling out easily
        - Sometimes, severe malnutrition may lead to unresponsiveness (stupor)
        - If calorie deficiency continues for long enough, there may be heart, liver and respiratory failure
        - Total starvation is said to be fatal within 8 to 12 weeks

- Causes of malnutrition are:
    - *Poor diet* - if a person does not eat enough food, or if what they eat does not provide them with the nutrients they require for good health, they suffer from malnutrition. Poor diet may be caused by one of several different factors. If the patient develops dysphagia (swallowing difficulties) because of an illness, or when recovering from an illness, they may not be able to consume enough of the right nutrients.

    - *Mental health problems* - some patients with mental health conditions, such as depression, may develop eating habits which lead to malnutrition. Patients with anorexia nervosa or bulimia may develop malnutrition because they are ingesting too little food.

    - *Mobility problems* - people with mobility problems may suffer from malnutrition simply because they either cannot get out enough to buy foods, or find preparing them too arduous.

    - *Digestive disorders and stomach conditions* - some people may eat properly, but their bodies cannot absorb the nutrients they need for good health.

    - *Alcoholism* - this is a chronic (long-term) disease. Individuals who suffer from alcoholism can develop gastritis, or pancreas damage. These problems also seriously undermine the body's ability to digest food, absorb certain vitamins, and produce hormones which regulate metabolism. Alcohol contains calories, reducing the patient's feeling of hunger, so he/she consequently may not eat enough proper food to supply the body with essential nutrients.

- Food shortage, price and distribution
- Lack of breastfeeding

Impact of Malnutrition on Health and Development

- Malnutrition and starvation have devastating impact on children, adults and especially on pregnant women. They also have severe and far-reaching socio-economic impacts.
  - *Effects on children* - Mental retardation, stunted growth, poor immune system, micronutrient deficiency, GI tract infections, anemia and inevitably – death.
  - *Effects on pregnant women* - Besides the health of the child being poor, a pregnant woman might have difficult labour, postpartum hemorrhage and anemia.
  - *General effects* - Weak immunity, inactivity of muscles, apathy, depression, and kidney function impairment.

- Malnourished children experience developmental delays, weight-loss and illness as a result of inadequate intake of protein, calories and other nutrients. Because orphaned and institutionalized children may experience one or several macronutrient and micronutrient deficiencies, they are at risk for a variety of short-term and long-term complications:

  - Short-Term Implications - Because so much development occurs in the first few years of life, nutrient deficiencies can have major short-term implications in young children.
    - **Immune Implications -** Malnourishment can greatly compromise a child's immune system, making them more susceptible to infectious diseases. Particularly in institutions where there are poor sanitary practices, children are vulnerable to infections from other children or caregivers.

    - **Growth Implications -** A child may contract an infection due to poor nutritional status. In turn, a gastrointestinal infection places the child at even greater risk for nutrient deficiencies because nutrients are unable to be absorbed properly. Consequently, nutrient deficiency combined with infection can cause growth retardation.

    - Additionally, a deficiency in one nutrient may lead to a deficiency in another nutrient. For example, deficiencies in iron, magnesium and zinc can cause anorexia and thereby result in reduced intake of other important nutrients such as protein. Low lipid intake can also affect the absorption of important fat-soluble vitamins such as vitamins A and Zinc and protein deficiencies can retard bone growth and development, putting a child at risk for long-term complications.

- Long-Term Implications - The short-term implications of malnutrition eventually give way to long-term complications, such as growth and cognitive delays:
    - Growth Implications - Malnutrition not only impacts growth in the short term, but can also limit total bone growth. Additionally, children classified as low height-for-age (stunted) may never be able to regain lost growth potential if they continue to live in a nutritionally deprived situation.
    - **Cognitive Implications -** Malnutrition negatively effects brain development causing delays in motor and cognitive development, such as:
        - Attention deficit disorder
        - Learning disabilities and impaired school performance
        - Decreased IQ scores
        - Memory deficiency
        - Reduced social skills
        - Reduced language development

********Field practice *********

Group1. Surveillance of food chain

1. What methods can you adopt in assessing food for possible infestation with germs, toxic?
2. Develop a food contamination monitoring questionnaire
3. Use the questionnaire to assess foodstuff in a local market

Group2. Nutritional epidemiology

1. Identify all the relationship between nutrition and health
2. Develop a food frequency assessment tool
3. Use the tool to assess the food frequency among pupils attending a local primary school

Group3. Clinical nutrition and diabetes

1. Develop a dummy table eliciting cases of diabetes by age and sex
2. Use the dummy table to capture the number of diabetes cases in a nearby hospital for 5 months
3. Randomize 10 cases of diabetes notes and assess presence of nutritional therapy

Group4. Nutrition education

1. Identify communication strategies that would be suitable in promoting adoption of food choice and nutritional behaviors
2. Use the communication strategy to sensitize 5 mothers with children having malnutrition related conditions in a pediatric ward in a nearby hospital
3. Use your mobile phone to take a picture of sensitization session going on

Group5. Food and nutrition policy

1. Visit a nearby hospital, identify nutrition office or officer, ask about all policy documents on nutrition
2. Make a tour in MCH clinic, medical ward and pediatric ward to identify availability of nutritional policy documents
3. Explain in plenary the content of the policy document to the class

Group6. Food industry and standard agency

1. Visit a nearby supermarket, sample 10 different foodstuff to identify the food industry that has manufactured it
2. Identify 10 foodstuff at the shelf and document all the ingredients that makes the foodstuff

- Identify if the food's safety has been certified by the Kenya Bureau of Standards.

Group7. Mass catering

1. Visit the Siaya MTC rationing store and identify different supplies that are catered
2. In each of the supply, identify the quantity and frequency of catering
3. Describe the documentation of the supplies on arrival till during usage